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**Software Requirements
Specification**

JPass: Java Password Manager

Version 1.0

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February 25, 2026

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1 Introduction

1.1 Purpose

JPass is a password manager that works on your computer and is built using Java. It helps you store all your passwords safely in one place. Each password is protected using AES-256 encryption, which keeps your information private and secure. With JPass, you only need to remember one master password to access all your accounts. It makes managing multiple passwords much easier. You can quickly add, edit, or delete your login details. The system also reduces the risk of using weak or repeated passwords. Overall, JPass simplifies password management while keeping your credentials safe. It is a useful tool for anyone who wants to stay secure online.

1.2 Document Conventions

All functional requirements are given unique numbers like REQ-1, REQ-2, and so on. Each requirement has a priority level marked as High, Medium, or Low. The steps of how the system should respond are written as Stimulus/Response sequences. The document is organized and formatted according to IEEE standards. This ensures clarity, traceability, and easy understanding for developers and reviewers.

1.3 Intended Audience and Reading Suggestions

This SRS is meant for developers, testers, project managers, and end-users. Developers use it to build the system correctly. Testers rely on it to check if the system works as expected. Project managers use it to plan and track the project. End-users can look at the main functions to see what the system can do. It helps everyone understand their role and the system's capabilities clearly.

1.4 Product Scope

JPass offers a secure password vault where users can store all their passwords in one place. It can generate strong passwords automatically and lets users search for stored credentials easily. The application supports importing and exporting password data and allows copying passwords to the clipboard safely. JPass works as a standalone desktop program and runs on Windows, Linux, and macOS. It is designed to simplify password management for users. In the future, it may add features like cloud backup and syncing passwords across multiple devices for more convenience.

1.5 References

- IEEE LaTeX SRS template <https://github.com/bonellia/srs-ieee-latex>
- Java Project References <https://github.com/gaborbata/jpass/tree/master>

2 Overall Description

2.1 Product Perspective

JPass is a modern, standalone application designed to replace manual password storage. It keeps all passwords in encrypted local vaults for strong security. Users no longer need to remember or write down multiple passwords. The system makes accessing and managing credentials simple and safe. In the future, JPass may offer optional cloud integration. This will allow users to access their passwords securely from multiple devices. It is built to be easy to use while keeping data private and protected at all times.

2.2 Product Functions

The system provides the following features:

1. Store credentials securely in an encrypted vault.
2. Update or delete existing credentials.
3. Generate strong, random passwords automatically.
4. Search for specific entries within the password vault.
5. Export encrypted backups of stored credentials.
6. Copy passwords safely to the clipboard for quick use.

2.3 User Classes and Characteristics

The users of JPass are individuals with basic desktop computer skills. They should understand how to use a master password to access their vault. Basic knowledge of file management is also needed, such as saving, opening, or organizing files. No advanced or specialized IT knowledge is required to use the system. It is designed to be simple and user-friendly for everyday users.

2.4 Operating Environment

The software is compatible with Java 11 or higher and runs on Windows, macOS, and Linux desktop systems. Users interact with the application using a standard keyboard and mouse. No special hardware or input devices are needed.

2.5 Design and Implementation Constraints

The system must use AES-256 encryption for all stored data. The graphical user interface (GUI) should be implemented using JavaFX or Swing. The master password must never be stored in plaintext under any circumstances. All vault operations, such as adding, retrieving, or updating credentials, should respond within 1 second to ensure smooth performance.

2.6 User Documentation

User manual, on-screen help, and tutorials will accompany the software. Documentation is provided in PDF format.

2.7 Assumptions and Dependencies

Users must have Java installed to run the application. If cloud features are used in the future, an active internet connection will be required. The overall performance of the system may vary depending on the hardware specifications of the user's computer.

3 External Interface Requirements

3.1 User Interfaces

GUI includes login, vault view, password generator, search, import/export, and clipboard management. Feedback messages guide users.

3.2 Hardware Interfaces

Standard desktop hardware. Minimum 4GB RAM, any display supporting 1024x768.

3.3 Software Interfaces

Java runtime, local file system for vault storage, optional HTTPS for cloud.

3.4 Communications Interfaces

Currently no network required. Future cloud sync will use secure HTTPS.

4 System Features

4.1 Master Password Authentication

4.1.1 Description and Priority

High priority. Ensures only authorized access.

4.1.2 Stimulus/Response Sequences

User enters master password → system validates → access granted or denied. System locks vault after multiple failed attempts.

4.1.3 Functional Requirements

REQ-1: System must allow creation of a master password.

REQ-2: System must deny access after 5 failed attempts.

REQ-3: System must never store master password in plaintext.

4.2 Password Storage

4.2.1 Description and Priority

High priority. Secure storage of all credentials.

4.2.2 Stimulus/Response Sequences

User adds/updates/deletes credentials → system encrypts → data saved in vault file.

4.2.3 Functional Requirements

REQ-4: System must encrypt all credentials using AES-256.

REQ-5: Users must be able to add, update, delete credentials.

REQ-6: System must maintain integrity of vault even if app closes unexpectedly.

4.3 Password Generation

4.3.1 Description and Priority

Medium priority. Helps users create strong passwords.

4.3.2 Stimulus/Response Sequences

User opens generator → selects length/complexity → system generates password → optionally saves to vault.

4.3.3 Functional Requirements

REQ-7: System must generate passwords with letters, numbers, symbols.

REQ-8: Users can configure length and complexity.

4.4 Password Retrieval and Search

4.4.1 Description and Priority

High priority. Efficiently retrieve credentials.

4.4.2 Stimulus/Response Sequences

User searches by username/URL → system returns matching credentials → allows copy to clipboard.

4.4.3 Functional Requirements

REQ-9: System must support search by username, account, or URL.

REQ-10: System must allow copying password to clipboard securely.

4.5 Vault Export/Backup

4.5.1 Description and Priority

Medium priority. Ensures data recovery.

4.5.2 Stimulus/Response Sequences

User exports vault → system encrypts → writes to selected location.

4.5.3 Functional Requirements

REQ-11: System must export encrypted vault.

REQ-12: Exported vault must be importable back to JPass.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

Vault operations, including adding, searching, or retrieving credentials, must complete within 1 second, even for large vaults. The system should remain fully responsive at all times, ensuring smooth and uninterrupted user interaction.

5.2 Safety Requirements

The system must ensure that user data is not lost in case of application crashes. Vault file corruption should be minimized through secure storage practices. Users must have options to create backups of their vault to protect against data loss. .

5.3 Security Requirements

The system must use AES-256 encryption to protect all stored passwords. A master password is required to access the vault. Measures must be in place to prevent any unauthorized access to user credentials.

5.4 Software Quality Attributes

The system must be reliable, maintainable, and easy to use. The graphical user interface (GUI) should be intuitive for users. The code should be modular, well-structured, and properly documented to simplify maintenance and future updates.

5.5 Business Rules

Only authenticated users are allowed to access their own vault. All vault operations, such as adding, editing, or deleting credentials, are restricted to the vault owner. Passwords copied to the clipboard must expire according to defined security rules to prevent misuse. .

6 Other Requirements

The system must document the database format, support for internationalization, and compliance with legal regulations. Any future requirements for cloud backup should also be included. A maintenance plan must be established to ensure the system remains secure, updated, and fully functional over time.

Glossary

Master Password

Vault

AES

GUI

Credential

Clipboard

To Be Determined List

Future cloud sync integration

Multi-device support

Biometric authentication